# OMER YOUSIF

Johor Bahru, Malaysia | +60 19-688 5468 | yousifali@graduate.utm.my

LinkedIn: linkedin.com/in/omer-yousif-174630331

## SUMMARY

Final-year Electrical Engineering student at UTM (CGPA 3.90/4.00) with 12 weeks of industrial training at a First-Grade engineering consultancy in Saudi Arabia, covering building power distribution, fire alarm and MEP systems. Skilled in load flow and voltage stability analysis (PowerWorld, ETAP), control system design (MATLAB) and renewable energy, with working knowledge of Python.

## EDUCATION

**Universiti Teknologi Malaysia (UTM)** **Johor Bahru, Malaysia**

*Bachelor of Electrical Engineering with Honours (Final Year)* Expected Graduation: Oct 2027

- **Current CGPA: 3.90/4.00** **Credit Hours Counted: 106**
- **Awards:** Pioneer's Academic Excellence Award (2024, 2025 & 2026), Dean's List Award (5 Semesters).

**Alsheikh Mustafa Alamein Secondary School** **Khartoum, Sudan**

*Scientific Field – Engineering Science* Score: 93.1%

## PROFESSIONAL EXPERIENCE

**Rawafd Alrwad Engineering Consultancy** **Buraidah, Saudi Arabia**

*Electrical Engineering Intern – Studies, Designs & Sustainability Department* Jul 2026 – Oct 2026

- Traced building power distribution from the utility supply through transformers, RMUs and ATS changeover to the MDB, EMDB and sub-distribution boards by matching single-line diagrams to installed switchgear on active sites.
- Inspected fire alarm and fire-fighting systems (addressable FACP loops, sprinkler flow and tamper switches, fire-pump MCC) and mapped their cause-and-effect interlocks with AHUs, smoke dampers and lifts.
- Reviewed as-built MEP coordination drawings against site conditions and punch lists; studied standby generators, lighting control panels and solar PV with battery backup for life-safety loads.
- Supported the company's qualification submission for the Jabal Umm Nuthaylah International Site Design Ideas Competition (Al-Madinah Region Development Authority): structured the master plan documentation, contributed dark-sky lighting concepts and wrote five precedent case studies.
- Compiled the five-member team CV package and assembled the final qualification package for supervisor review.
- Completed self-directed study of Python programming (MITx 6.00.1x) and neural network fundamentals, writing tested programs that model electrical loads and components.

## ACADEMIC PROJECTS: CONTROL SYSTEMS & AUTOMATION

**Control System Design & Stability Analysis** **Feb 2026**

*MATLAB, Root Locus, Bode Plots, Time/Frequency Domain Analysis*

- Designed proportional controllers to meet 10% and 20% overshoot targets for second- and third-order plants, cross-checking the gain with MATLAB iteration, analytical damping-ratio design and Root Locus (all three agreed within 0.6% for the 10% target).
- Extracted Gain Margin (GM), Phase Margin (PM) and crossover frequencies from Bode plots of three open-loop systems and compared their robustness.

- Identified each system type and calculated the steady-state error constants ($K_p$, $K_v$, $K_a$) to characterise performance limits.

**Self-Balancing Robot** **Apr 2025 – May 2025**
*ESP32-C3, PID Control, KiCad, SolidWorks*

- Engineered a self-balancing robot, experimentally tuning PID control parameters to achieve real-time stability on an ESP32-C3 microcontroller.
- Designed custom PCBs in KiCad and the mechanical chassis in SolidWorks; integrated an MPU6050 inertial sensor via I2C.

## ACADEMIC PROJECTS: POWER SYSTEMS & RENEWABLE ENERGY

**Load Flow & Voltage Stability Analysis (IEEE 14-Bus System)** **Jan 2026**
*PowerWorld Simulator, Power Systems Analysis*

- Modelled the 132 kV IEEE 14-bus network in PowerWorld and found every PV generator at its reactive power limit, leaving Bus 14 the weakest bus at 0.913 p.u.
- Simulated a 150% load stress case, in which Bus 14 collapsed to 0.646 p.u. and 10 of the 14 buses fell below 0.80 p.u.
- Raised the Bus 14 voltage by 5.7% with a 9 Mvar shunt capacitor and by 5.3% by increasing the Bus 8 generator's $Q_{max}$, and compared both methods on voltage support and system losses.

**Grid-Connected Solar PV Design Workshop** **Dec 2024**
*Faculty of Electrical Engineering, UTM*

- Calculated Peak Sun Hours (PSH) and module efficiency to size Grid-Connected PV (GCPV) systems.
- Installed PV modules and MC4 connectors hands-on, following high-voltage DC safety practice.

## ACADEMIC PROJECTS: HARDWARE & APPLIED PHYSICS

**Electromagnetic Field Strength Optimization** **Jan 2025**
*Experimental Design, Data Analysis, Hardware Assembly*

- Designed and ran an experiment quantifying how coil turns, applied voltage, wire diameter and core material affect electromagnet strength.
- Built multiple electromagnet configurations with copper windings, different core materials and DC sources, measuring strength by lifting capacity.
- Lead author of the group's journal-format technical report identifying the configuration that gives maximum magnetic strength.

**Automatic Safe System Design** **Jan 2024**
*ATmega32, C Programming, Microchip Studio, Proteus VSM*

- Programmed an ATmega32 safe with 4-digit password logic, a keypad, a 7-segment countdown display, status LEDs and servo-driven door control; validated in Proteus VSM.

## LEADERSHIP, ACTIVITIES & VOLUNTEERING

**Industrial Technical Visit | Sunway Paving Solutions** **Dec 2025**
*Focus: Net Zero Carbon & Green Manufacturing*

- Observed the industrial implementation of Net Zero Carbon goals and a 2,327 MWh solar PV system for manufacturing.

**Judge | International Youth Robot Competition (IYRC) ASEAN** **Jun 2025**
*Theme: Green Energy & Sustainability*

- Evaluated robotics projects from 800+ young innovators, assessing technical logic and sustainable design solutions.

**UTM Green DNA** **Oct 2024 – Present**

*Member | Sustainability & Green Education*

- Support UTM's goal of becoming a Green Hub by promoting Carbon Neutral and Zero Waste initiatives on campus.
- Take part in capacity-building activities that foster sustainable development and a circular-economy mindset among students.

**The Institution of Engineers, Malaysia (IEM)** **Nov 2025 – Present**

*Student Member (Membership No: 133388)*

- Attend engineering seminars and industrial networking events.

**Wellness Charity** **Mar 2020 – 2022**

*General Volunteer | Khartoum, Sudan*

- Managed legal compliance documentation and streamlined organisational workflows to support secretariat functions.

**Steering Engineering Society (University of Khartoum)** **May 2020 – 2021**

*Public Relations & Career Fair Team*

- Coordinated the annual Career Fair and secured sponsorship funding through institutional visits.
- Managed financial reports and used Excel for budget tracking and event logistics.

## CERTIFICATIONS & COURSES

**Preparation for LEED Green Associate Certification** **In Progress**

*Professional Development*

- Studying sustainable building strategies, energy efficiency and LEED rating systems.

## TECHNICAL SKILLS

- **Power & Renewable Energy:** Load Flow Analysis, Voltage Stability, Reactive Power Compensation, Grid-Connected PV Sizing, PSH Analysis, LEED Principles.
- **Building Services (MEP):** LV Distribution (MDB, EMDB, SMDB), Single-Line Diagram Reading, Fire Alarm Systems & Cause-and-Effect Logic, Emergency & Standby Power, Saudi Building Code (SBC 401), NFPA 72.
- **Control Systems & Electronics:** PID Control, Root Locus, Bode Plots, ESP32, ATmega32, Arduino Microcontrollers.
- **Engineering Software:** MATLAB, PowerWorld Simulator, ETAP, AutoCAD, SolidWorks, KiCad, Proteus VSM, Microchip Studio.
- **Programming Languages:** Python, C, C++.
- **Languages:** English, Arabic.